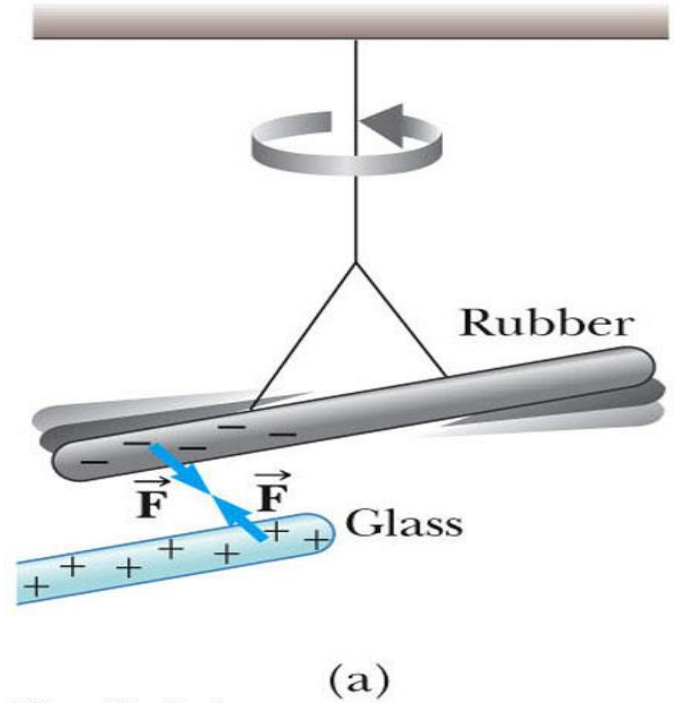
Chapter 23

Electric Fields

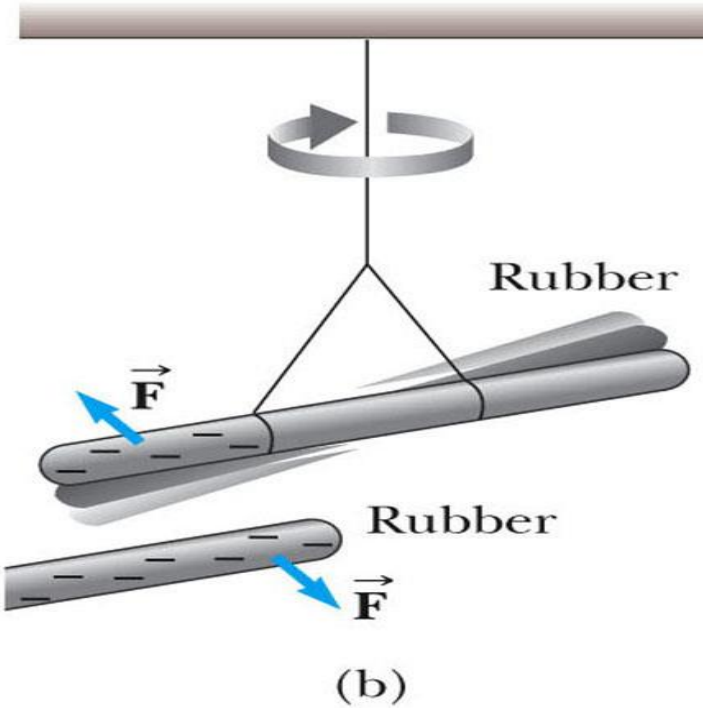
Sec 23.1: Properties of Electric Charges

- There are two kinds of electric charges
  - Called positive and negative
    - Negative charges are the type possessed by electrons
    - Positive charges are the type possessed by protons
- Charges of the same sign repel one another and charges with opposite signs attract one another
- Electric charge is always conserved in an isolated system
  - For example, the charge is not created in the process of rubbing two objects together
  - The electrification is due to a transfer of charge from one object to another

- The rubber rod is negatively charged
- The glass rod is positively charged
- The two rods will attract

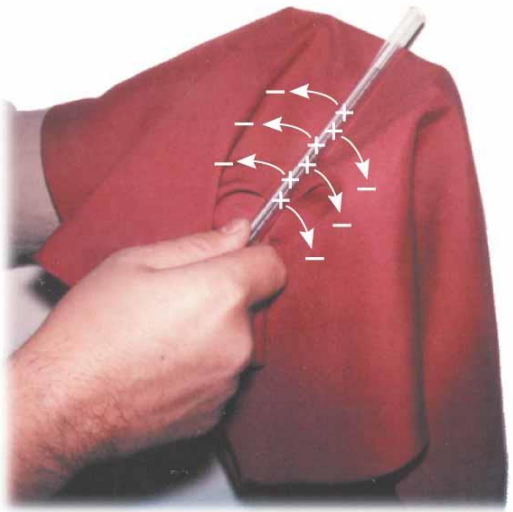


- The rubber rod is negatively charged
- The second rubber rod is also negatively charged
- The two rods will repel



Conservation of Electric Charges

- A glass rod is rubbed with silk, Electrons are transferred from the glass to the silk
- Each electron adds a negative charge to the silk, An equal positive charge is left on the rod



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Quantization of Electric Charges

- The electric charge, q, is said to be quantized
  - *q* is the standard symbol used for charge as a variable
  - Electric charge exists as discrete packets
  - $q = \pm Ne$
  - *N is an integer, e is the fundamental unit of charge,* $|e| = 1.6 \times 10^{-19} C$
  - *Electron:* $q = -e$, *Proton:* $q = +e$

Quick Quiz 23.1 Three objects are brought close to each other, two at a time. When objects A and B are brought together, they repel. When objects B and C are brought together, they also repel. Which of the following are true? **(a)** Objects A and C possess charges of the same sign. **(b)** Objects A and C possess charges of opposite sign. **(c)** All three objects possess charges of the same sign. **(d)** One object is neutral. **(e)** Additional experiments must be performed to determine the signs of the charges.

Sec 23.2: Charging Objects by Induction: Conductors

- Electrical conductors are materials in which some of the electrons are free electrons
  - Free electrons are not bound to the atoms
  - These electrons can move relatively freely through the material
  - Examples of good conductors include copper, aluminum, and silver (metals).
  - When a good conductor is charged in a small region, the charge readily distributes itself over the entire surface of the material

Insulators

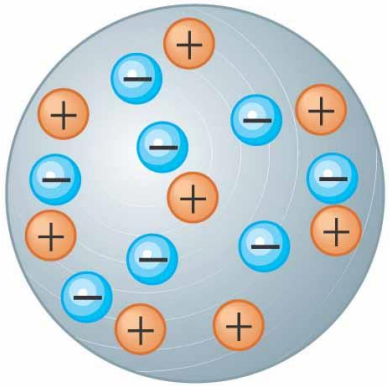
- Electrical insulators are materials in which all of the electrons are bound to atoms
  - These electrons can not move relatively freely through the material
  - Examples of good insulators include glass, rubber, and wood
  - When a good insulator is charged in a small region, the charge is unable to move to other regions of the material

Semiconductors

- The electrical properties of semiconductors are somewhere between those of insulators and conductors
- Examples of semiconductor materials include silicon and germanium

Charging by Induction

- Charging by induction requires no contact with the object inducing the charge
- Assume we start with a neutral metallic sphere
  - The sphere has the same number of positive and negative charges



(a)

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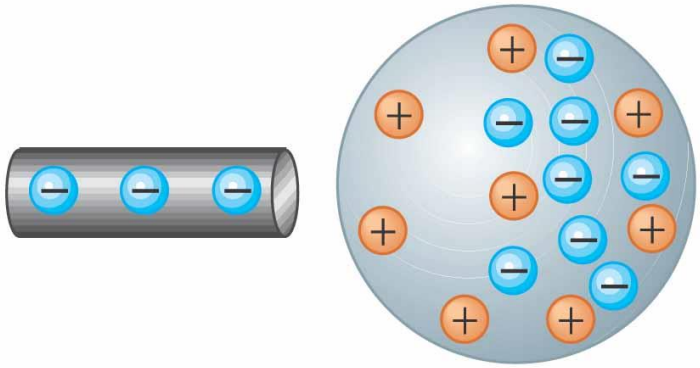
- A charged rubber rod is placed near the sphere
  - It does **not** touch the sphere
- The electrons in the neutral sphere are redistributed



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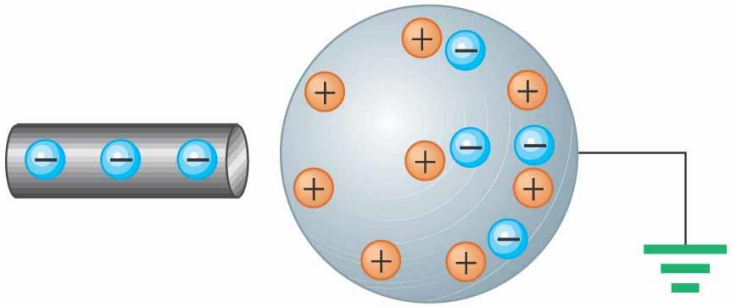
- The sphere is grounded
- Some electrons can leave the sphere through the ground wire



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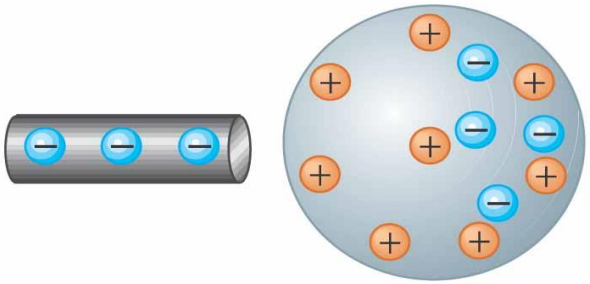
- The ground wire is removed
- There will now be more positive charges
- The charges are not uniformly distributed
- The positive charge has been ***induced*** in the sphere



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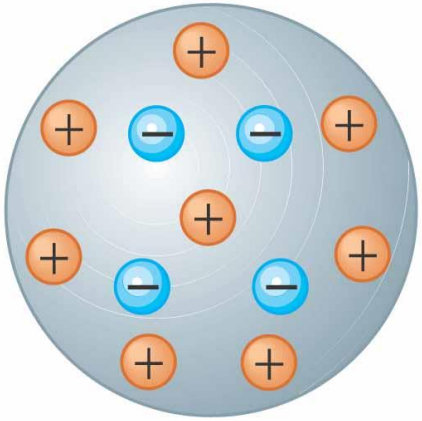
- The rod is removed
- The electrons remaining on the sphere redistribute themselves
- There is still a net positive charge on the sphere
- The charge is now uniformly distributed



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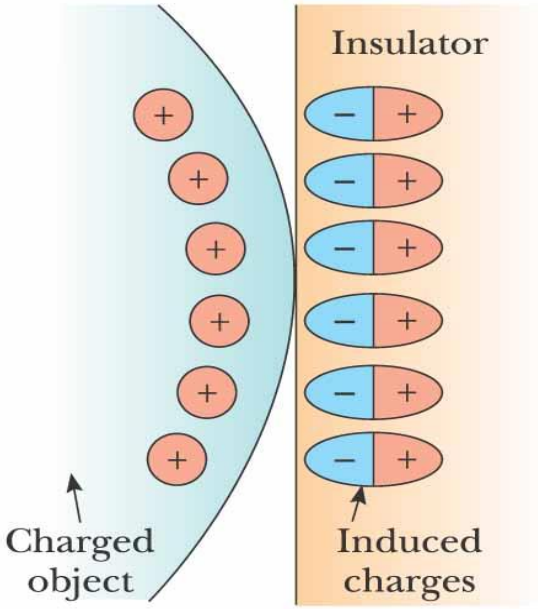
Charge Rearrangement in Insulators

- A process similar to induction can take place in insulators
- The charges within the molecules of the material are rearranged



(a)

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Quick Quiz 23.2 Three objects are brought close to one another, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. Which of the following are necessarily true? **(a)** Objects A and C possess charges of the same sign. **(b)** Objects A and C possess charges of opposite sign. **(c)** All three objects possess charges of the same sign. **(d)** One object is neutral. **(e)** Additional experiments must be performed to determine information about the charges on the objects.

Sec 23.3: Coulomb's Law :

- Charles Coulomb measured the magnitudes of electric forces between two small charged spheres
- He found the force depended on the charges and the distance between them
- The term **point charge** refers to a particle of zero sizes that carries an electric charge
  - The electrical behavior of electrons and protons is well described by modeling them as point charges
- Coulomb's Law gives the electrical force between two stationary point charges
- The force is inversely proportional to the square of the separation r between the charges and directed along the line joining them
- The force is proportional to the product of the charges, q1, and q2, on the two particles
- The force is attractive if the charges are of opposite sign, and is repulsive if the charges are of like sign

- The force is a conservative force
- Mathematically, $$F_e = k_e \frac{|q_1||q_2|}{r^2}$$
- The SI unit of charge is the coulomb (C)
- k_e is called the Coulomb constant
  - $k_e = 8.9876 \times 10^9\ \mathrm{N{\cdot}m^2/C^2} = 1/(4\pi\varepsilon_o)$
  - ε_o is the **permittivity of free space**
  - $\varepsilon_o = 8.8542 \times 10^{-12}\ \mathrm{C^2\ /\ N{\cdot}m^2}$
- Remember the charges need to be in coulombs
  - e is the smallest unit of charge
    - except quarks
  - $e = 1.6 \times 10^{-19}$ C
  - So 1 C needs 6.24×10^{18} electrons or protons
- Typical charges can be in the μC range
- Remember that force is a vector quantity

TABLE 23.1

Charge and Mass of the Electron, Proton, and Neutron

Particle	Charge (C)	Mass (kg)
Electron (e)	$-1.602\,176\,5 \times 10^{-19}$	$9.109\,4 \times 10^{-31}$
Proton (p)	$+1.602\,176\,5 \times 10^{-19}$	$1.672\,62 \times 10^{-27}$
Neutron (n)	0	$1.674\,93 \times 10^{-27}$

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Example 23.1 The Hydrogen Atom

The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately 5.3×10^{-11} m. Find the magnitudes of the electric force and the gravitational force between the two particles.

$$F_e = k_e \frac{|e||-e|}{r^2} = (8.988 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(1.60 \times 10^{-19} \text{ C})^2}{(5.3 \times 10^{-11} \text{ m})^2}$$

$$= \boxed{8.2 \times 10^{-8} \text{ N}}$$

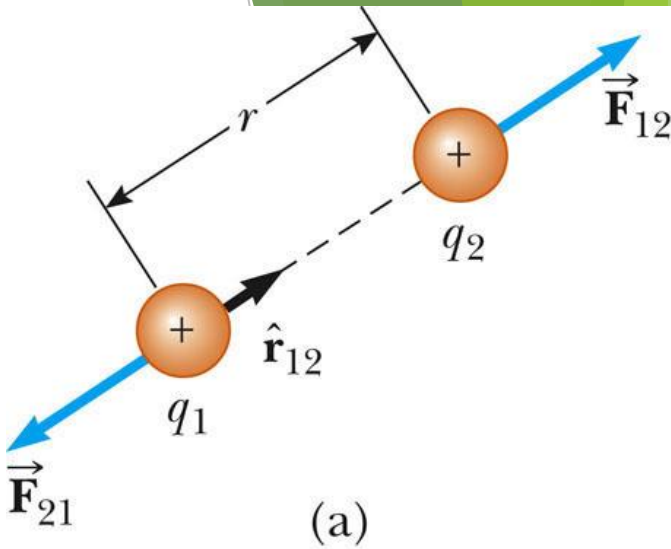
$$F_g = G \frac{m_e m_p}{r^2}$$

$$= (6.674 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2) \frac{(9.11 \times 10^{-31} \text{ kg})(1.67 \times 10^{-27} \text{ kg})}{(5.3 \times 10^{-11} \text{ m})^2}$$

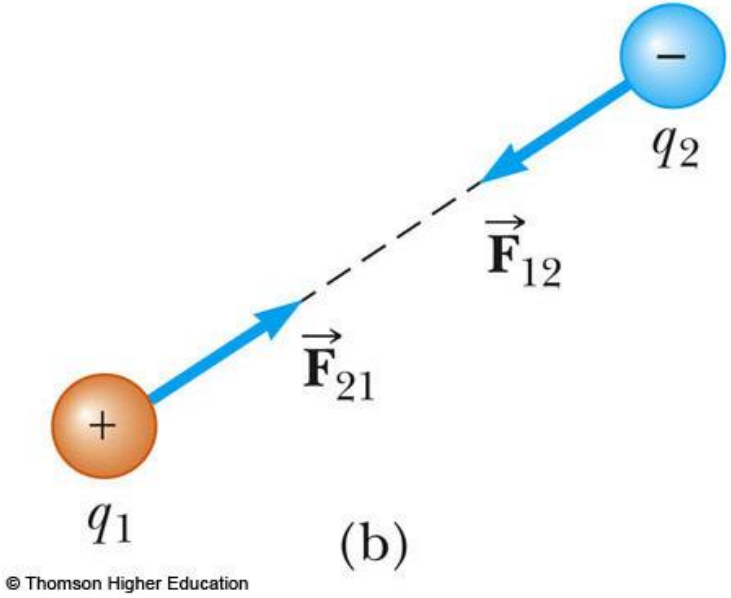
$$= \boxed{3.6 \times 10^{-47} \text{ N}}$$

Vector Nature of Electric Forces

- In vector form, $$\vec{\mathbf{F}}_{12} = k_e \frac{q_1 q_2}{r^2} \hat{\mathbf{r}}_{12}$$
- $\hat{\mathbf{r}}_{12}$ is a unit vector directed from q_1 to q_2
- The like charges produce a repulsive force between them
- Electrical forces obey Newton's Third Law
- The force on q1 is equal in magnitude and opposite in direction to the force on q2
  - $\vec{\mathbf{F}}_{21} = -\vec{\mathbf{F}}_{12}$
- With like signs for the charges, the product q1q2 is positive and the force is repulsive



- Two point charges are separated by a distance *r*
- The unlike charges produce an attractive force between them
- With unlike signs for the charges, the product q_1q_2 is negative and the force is attractive
- The sign of the product of q1q2 gives the relative direction of the force between q1 and q2
- The absolute direction is determined by the actual location of the charges



The Superposition Principle

- The resultant force on anyone charge equals the vector sum of the forces exerted by the other individual charges that are present
  - Remember to add the forces *as vectors*
- The resultant force on q_1 is the vector sum of all the forces exerted on it by other charges:

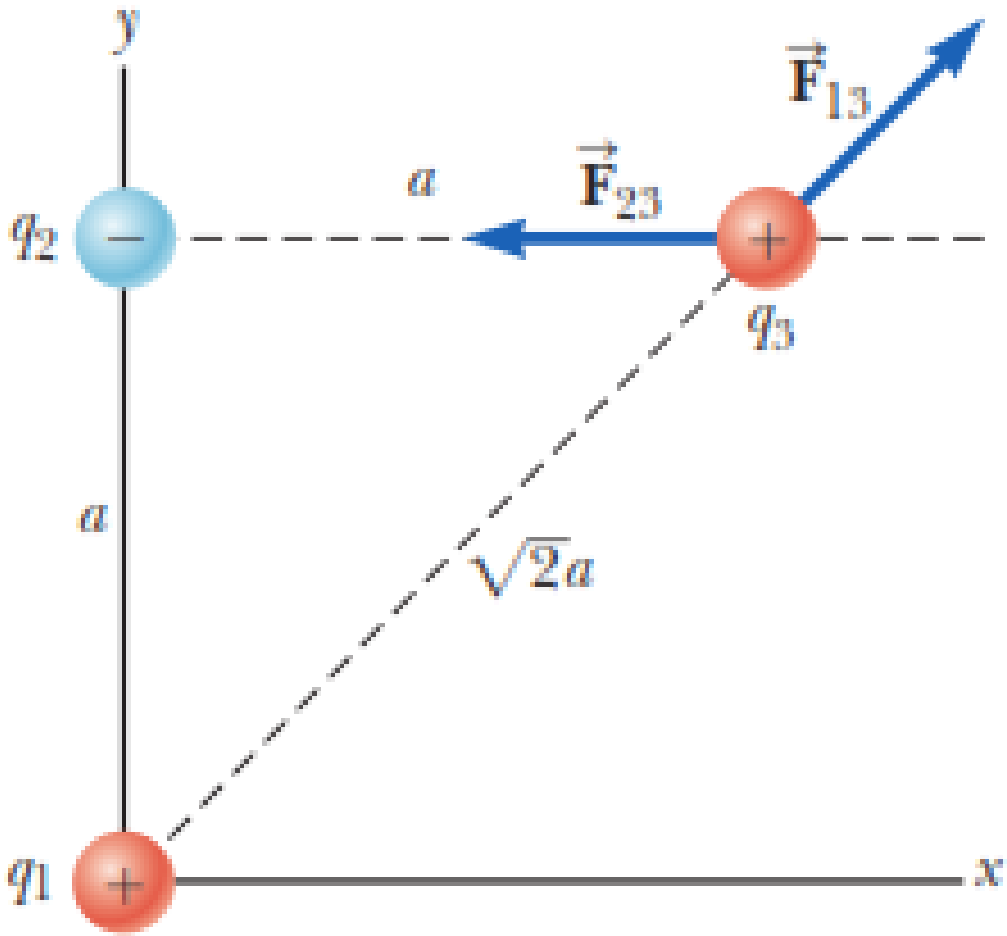
$$\vec{\mathbf{F}}_1 = \vec{\mathbf{F}}_{21} + \vec{\mathbf{F}}_{31} + \vec{\mathbf{F}}_{41}$$

Quick Quiz 23.3 Object A has a charge of $+2\ \mu C$, and object B has a charge of $+6\ \mu C$. Which statement is true about the electric forces on the objects?

(a) $\vec{\mathbf{F}}_{AB} = -3\vec{\mathbf{F}}_{BA}$ **(b)** $\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}$ **(c)** $3\vec{\mathbf{F}}_{AB} = -\vec{\mathbf{F}}_{BA}$ **(d)** $\vec{\mathbf{F}}_{AB} = 3\vec{\mathbf{F}}_{BA}$
(e) $\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$ **(f)** $3\vec{\mathbf{F}}_{AB} = \vec{\mathbf{F}}_{BA}$

Example 23.2 Find the Resultant Force

Consider three point charges located at the corners of a right triangle as shown in Figure 23.7, where $q_1 = q_3 = 5.00\ \mu\text{C}$, $q_2 = -2.00\ \mu\text{C}$, and $a = 0.100$ m. Find the resultant force exerted on q_3.



$$F_{23} = k_e \frac{|q_2||q_3|}{a^2}$$

$$= (8.988 \times 10^9\ \mathrm{N \cdot m^2/C^2}) \frac{(2.00 \times 10^{-6}\ \mathrm{C})(5.00 \times 10^{-6}\ \mathrm{C})}{(0.100\ \mathrm{m})^2} = 8.99\ \mathrm{N}$$

$$F_{13} = k_e \frac{|q_1||q_3|}{(\sqrt{2}\ a)^2}$$

$$= (8.988 \times 10^9\ \mathrm{N \cdot m^2/C^2}) \frac{(5.00 \times 10^{-6}\ \mathrm{C})(5.00 \times 10^{-6}\ \mathrm{C})}{2(0.100\ \mathrm{m})^2} = 11.2\ \mathrm{N}$$

$$F_{13x} = (11.2\ \mathrm{N}) \cos 45.0° = 7.94\ \mathrm{N}$$

$$F_{13y} = (11.2\ \mathrm{N}) \sin 45.0° = 7.94\ \mathrm{N}$$

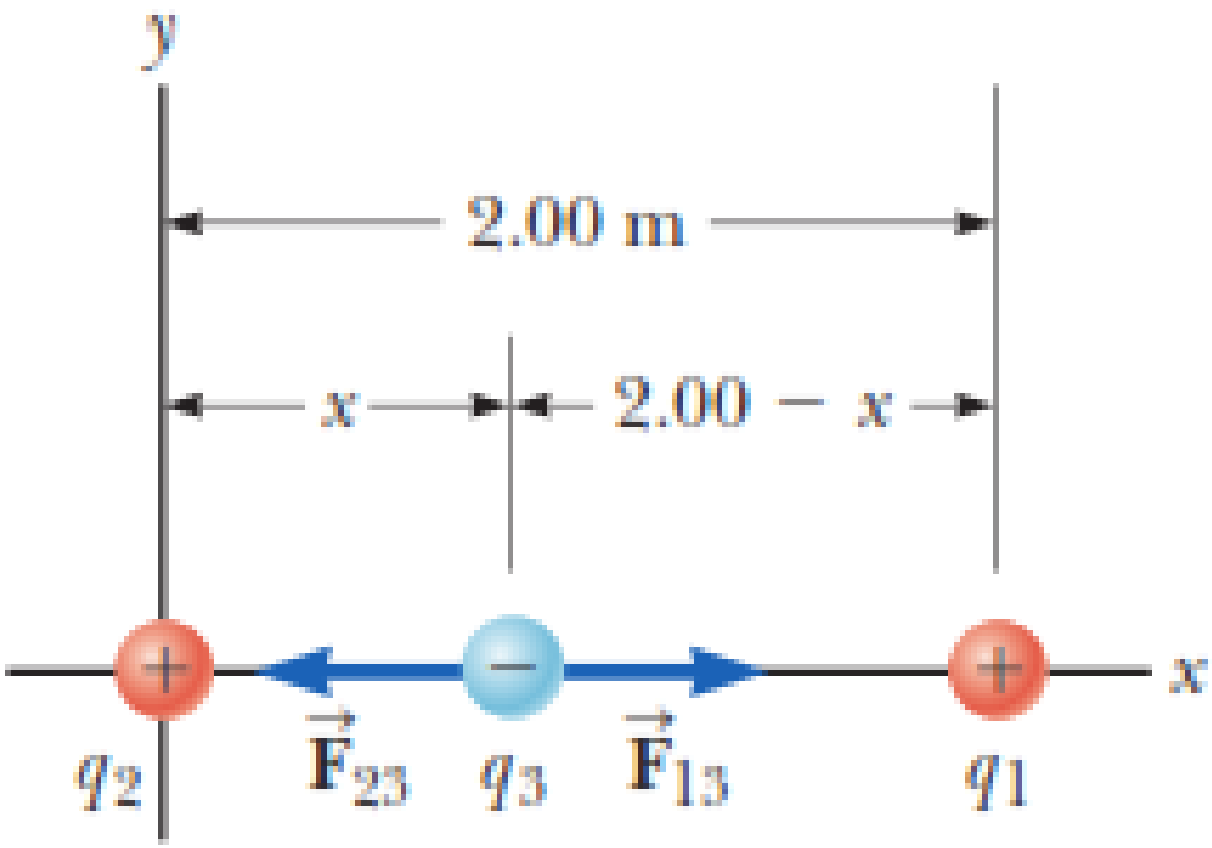
$$F_{3x} = F_{13x} + F_{23x} = 7.94\ \mathrm{N} + (-8.99\ \mathrm{N}) = -1.04\ \mathrm{N}$$

$$F_{3y} = F_{13y} + F_{23y} = 7.94\ \mathrm{N} + 0 = 7.94\ \mathrm{N}$$

$$\vec{\mathbf{F}}_3 = (-1.04\hat{\mathbf{i}} + 7.94\hat{\mathbf{j}})\ \mathrm{N}$$

Example 23.3 Where Is the Net Force Zero? AM

Three point charges lie along the x axis as shown in Figure 23.8. The positive charge $q_1 = 15.0\ \mu C$ is at $x = 2.00$ m, the positive charge $q_2 = 6.00\ \mu C$ is at the origin, and the net force acting on q_3 is zero. What is the x coordinate of q_3?



$$\vec{\mathbf{F}}_3 = \vec{\mathbf{F}}_{23} + \vec{\mathbf{F}}_{13} = -k_e \frac{|q_2||q_3|}{x^2}\,\hat{\mathbf{i}} + k_e \frac{|q_1||q_3|}{(2.00 - x)^2}\,\hat{\mathbf{i}} = 0$$

$$k_e \frac{|q_2||q_3|}{x^2} = k_e \frac{|q_1||q_3|}{(2.00 - x)^2}$$

$$(2.00 - x)^2|q_2| = x^2|q_1|$$

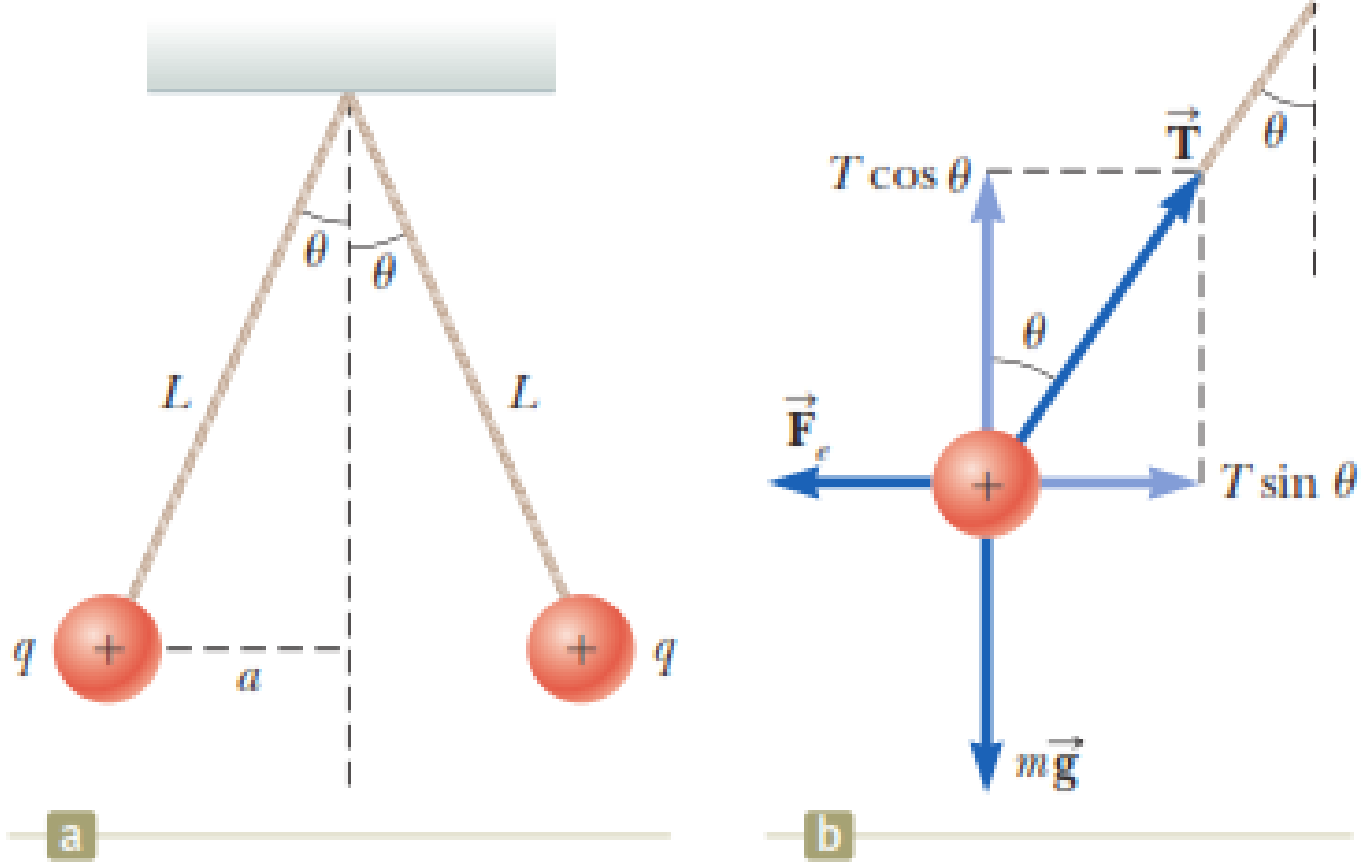
$$(2.00 - x)\sqrt{|q_2|} = \pm x\sqrt{|q_1|}$$

$$x = \frac{2.00\sqrt{|q_2|}}{\sqrt{|q_2|} \pm \sqrt{|q_1|}}$$

$$x = \frac{2.00\sqrt{6.00 \times 10^{-6}\,\mathrm{C}}}{\sqrt{6.00 \times 10^{-6}\,\mathrm{C}} + \sqrt{15.0 \times 10^{-6}\,\mathrm{C}}} = 0.775\ \mathrm{m}$$

Example 23.4 Find the Charge on the Spheres

Two identical small charged spheres, each having a mass of 3.00×10^{-2} kg, hang in equilibrium as shown in Figure 23.9a. The length L of each string is 0.150 m, and the angle θ is 5.00°. Find the magnitude of the charge on each sphere.



$$(1) \quad \sum F_x = T\sin\theta - F_e = 0 \rightarrow T\sin\theta = F_e$$

$$(2) \quad \sum F_y = T\cos\theta - mg = 0 \rightarrow T\cos\theta = mg$$

$$(3) \quad \tan\theta = \frac{F_e}{mg} \rightarrow F_e = mg\tan\theta$$

$$\sin\theta = \frac{a}{L} \rightarrow a = L\sin\theta$$

$$|q| = \sqrt{\frac{F_e r^2}{k_e}} = \sqrt{\frac{F_e(2a)^2}{k_e}} = \sqrt{\frac{mg\tan\theta(2L\sin\theta)^2}{k_e}}$$

$$|q| = \sqrt{\frac{(3.00\times10^{-2}\ \text{kg})(9.80\ \text{m/s}^2)\tan(5.00°)[2(0.150\ \text{m})\sin(5.00°)]^2}{8.988\times10^9\ \text{N}\cdot\text{m}^2/\text{C}^2}}$$

$$= \boxed{4.42\times10^{-8}\ \text{C}}$$

Sec 23.4: Analysis Model: Particle in a Field (Electric)

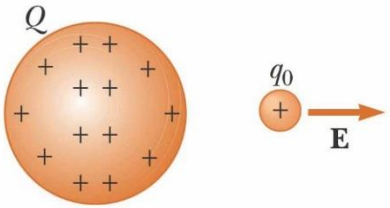
* Electric Field – Introduction

- The electric force is a field force
- Field forces can act through space
  - The effect is produced even with no physical contact between objects
- Faraday developed the concept of a field in terms of electric fields
- An **electric field** is said to exist in the region of space around a charged object
  - **This charged object is the source charge**
- When another charged object, the **test charge**, enters this electric field, an electric force acts on it
- The electric field is defined as the electric force on the test charge per unit charge

- The electric field vector, $\vec{\mathbf{E}}$, at a point in space is defined as the electric force $\vec{\mathbf{F}}$ acting on a positive test charge, q_o placed at that point divided by the test charge:

$$\vec{\mathbf{E}} \equiv \frac{\vec{\mathbf{F}}}{q_o}$$

- $\vec{\mathbf{E}}$ is the field produced by some charge or charge distribution, separate from the test charge
- The existence of an electric field is a property of the source charge
  - The presence of the test charge is not necessary for the field to exist
- The test charge serves as a detector of the field
- The direction of $\vec{\mathbf{E}}$ is that of the force on a positive test charge
- The SI units of $\vec{\mathbf{E}}$ are N/C
- We can also say that an electric field exists at a point if a test charge at that point experiences an electric force



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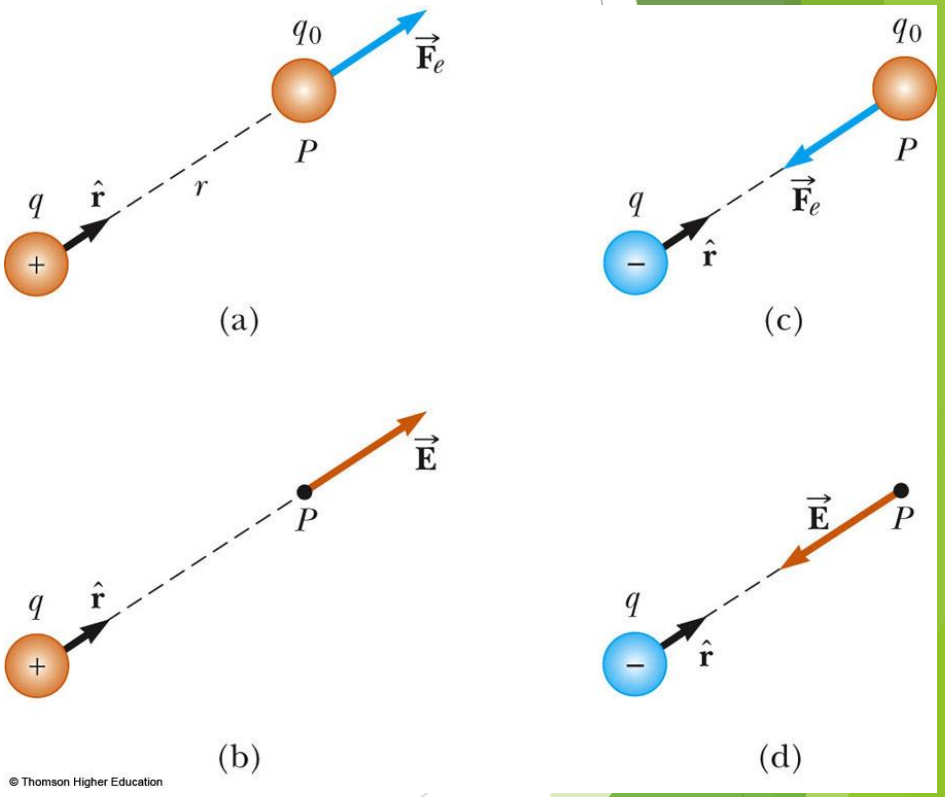
Relationship Between F and E

- $$\vec{\mathbf{F}}_e = q\vec{\mathbf{E}}$$
  - This is valid for a point charge only
  - One of zero size
  - For larger objects, the field may vary over the size of the object
- If *q* is positive, the force and the field are in the same direction
- If *q* is negative, the force and the field are in opposite directions
- Remember Coulomb's law, between the source and test charges, can be expressed as $$\vec{\mathbf{F}}_e = k_e \frac{q q_o}{r^2}\hat{\mathbf{r}}$$
- Then, the electric field will be $$\vec{\mathbf{E}} = \frac{\vec{\mathbf{F}}_e}{q_o} = k_e \frac{q}{r^2}\hat{\mathbf{r}}$$
- At any point *P*, the total electric field due to a group of source charges equals the vector sum of the electric fields of all the charges
- $$\vec{\mathbf{E}} = k_e \sum_i \frac{q_i}{r_i^2}\hat{\mathbf{r}}_i$$

More About Electric Field Direction

- a) *q* is positive, the force is directed away from *q*
- b) The direction of the field is also away from the positive source charge
- c) *q* is negative, the force is directed toward *q*
- d) The field is also toward the negative source charge



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Example 23.5 A Suspended Water Droplet AM

A water droplet of mass 3.00×10^{-12} kg is located in the air near the ground during a stormy day. An atmospheric electric field of magnitude 6.00×10^3 N/C points vertically downward in the vicinity of the water droplet. The droplet remains suspended at rest in the air. What is the electric charge on the droplet?

$$\sum F_y = 0 \rightarrow F_e - F_g = 0$$

$$q(-E) - mg = 0$$

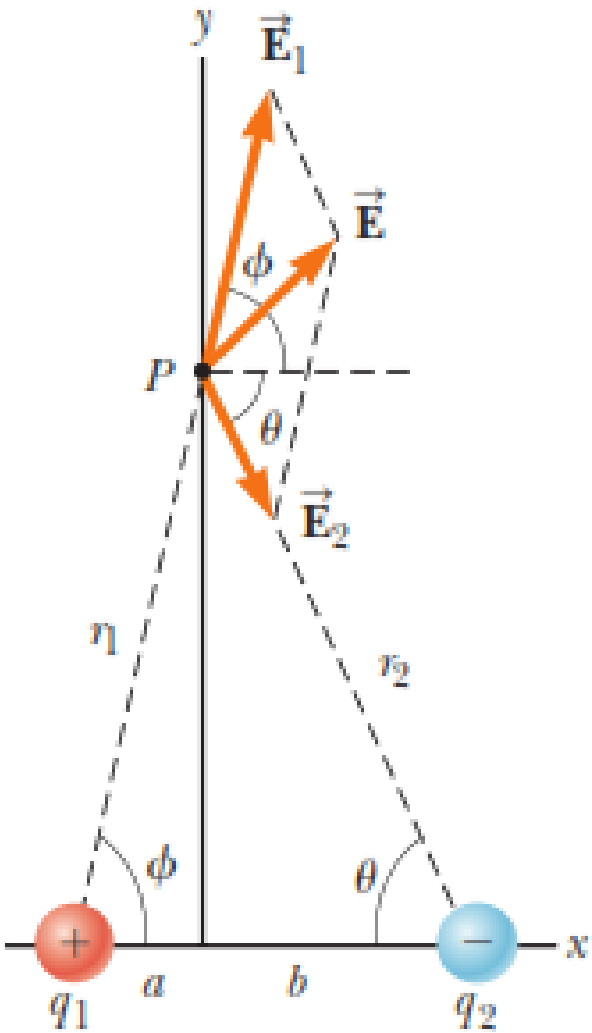
$$q = -\frac{mg}{E}$$

$$q = -\frac{(3.00 \times 10^{-12}\,\text{kg})(9.80\,\text{m/s}^2)}{6.00 \times 10^3\,\text{N/C}} = -4.90 \times 10^{-15}\,\text{C}$$

Example 23.6 Electric Field Due to Two Charges

Charges q_1 and q_2 are located on the x axis, at distances a and b, respectively, from the origin as shown in Figure 23.12.

(A) Find the components of the net electric field at the point P, which is at position $(0, y)$.



$$E_1 = k_e \frac{|q_1|}{r_1^2} = k_e \frac{|q_1|}{a^2 + y^2}$$

$$E_2 = k_e \frac{|q_2|}{r_2^2} = k_e \frac{|q_2|}{b^2 + y^2}$$

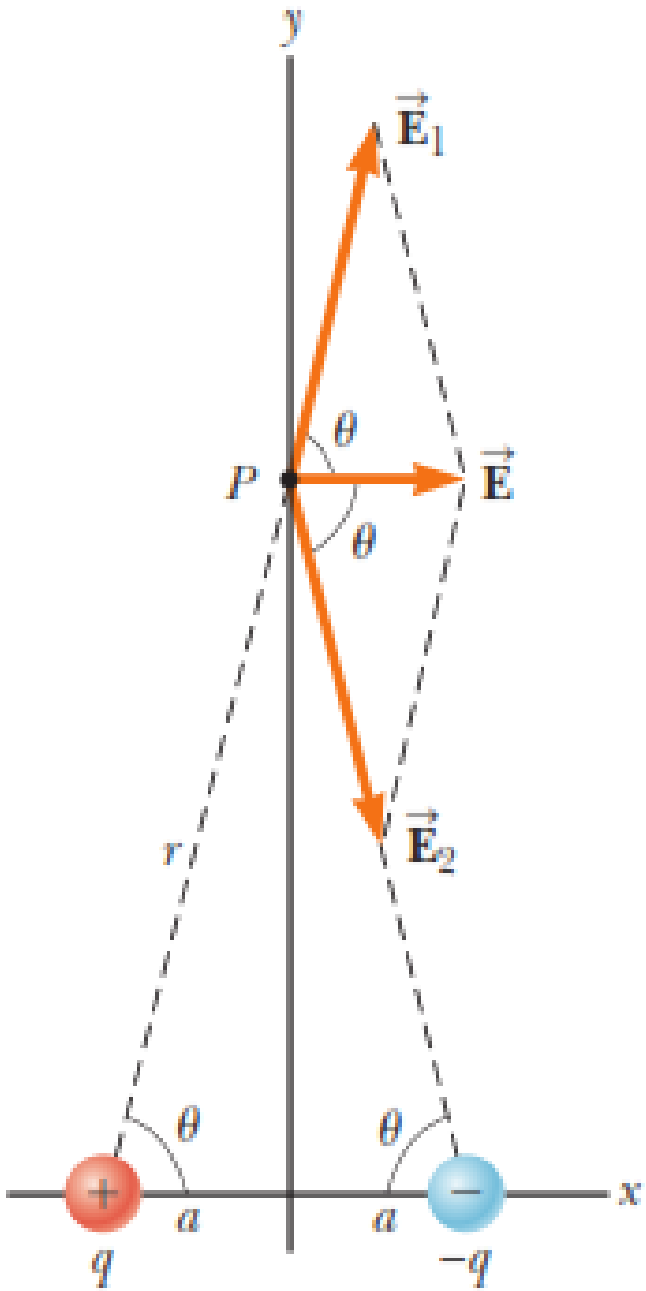
$$\vec{\mathbf{E}}_1 = k_e \frac{|q_1|}{a^2 + y^2} \cos\phi\,\hat{\mathbf{i}} + k_e \frac{|q_1|}{a^2 + y^2} \sin\phi\,\hat{\mathbf{j}}$$

$$\vec{\mathbf{E}}_2 = k_e \frac{|q_2|}{b^2 + y^2} \cos\theta\,\hat{\mathbf{i}} - k_e \frac{|q_2|}{b^2 + y^2} \sin\theta\,\hat{\mathbf{j}}$$

$$(1) \quad E_x = E_{1x} + E_{2x} = k_e \frac{|q_1|}{a^2 + y^2} \cos\phi + k_e \frac{|q_2|}{b^2 + y^2} \cos\theta$$

$$(2) \quad E_y = E_{1y} + E_{2y} = k_e \frac{|q_1|}{a^2 + y^2} \sin\phi - k_e \frac{|q_2|}{b^2 + y^2} \sin\theta$$

(B) Evaluate the electric field at point P in the special case that $|q_1| = |q_2|$ and $a = b$.



$$E_x = k_e \frac{q}{a^2 + y^2} \cos\theta + k_e \frac{q}{a^2 + y^2} \cos\theta = 2k_e \frac{q}{a^2 + y^2} \cos\theta$$

$$E_y = k_e \frac{q}{a^2 + y^2} \sin\theta - k_e \frac{q}{a^2 + y^2} \sin\theta = 0$$

$$\cos\theta = \frac{a}{r} = \frac{a}{(a^2 + y^2)^{1/2}}$$

$$E_x = 2k_e \frac{q}{a^2 + y^2} \left[\frac{a}{(a^2 + y^2)^{1/2}} \right] = k_e \frac{2aq}{(a^2 + y^2)^{3/2}}$$

(C) Find the electric field due to the electric dipole when point P is a distance $y >> a$ from the origin.

In the solution to part (B), because $y >> a$, neglect a^2 compared with y^2 and write the expression for E in this case:

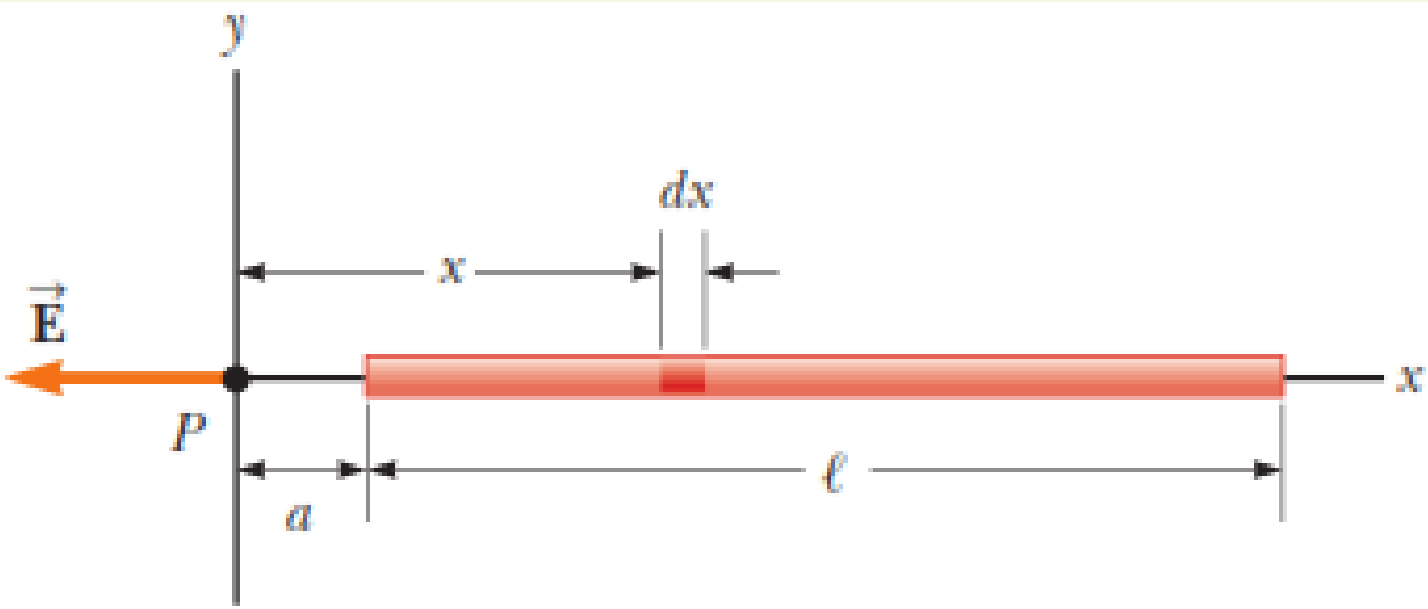
$$(5) \quad E \approx k_e \frac{2aq}{y^3}$$

23.5 Electric Field of a Continuous Charge Distribution

- The distances between charges in a group of charges may be much smaller than the distance between the group and a point of interest
- In this situation, the system of charges can be modeled as continuous
- The system of closely spaced charges is equivalent to a total charge that is continuously distributed along some line, over some surface, or throughout some volume
- **Volume charge density**: when a charge is distributed evenly throughout a volume
  - $\rho \equiv Q / V$ with units C/m^3
- **Surface charge density**: when a charge is distributed evenly over a surface area
  - $\sigma \equiv Q / A$ with units C/m^2
- **Linear charge density**: when a charge is distributed along a line
  - $\lambda \equiv Q / \ell$ with units C/m

Example 23.7 The Electric Field Due to a Charged Rod

A rod of length ℓ has a uniform positive charge per unit length λ and a total charge Q. Calculate the electric field at a point P that is located along the long axis of the rod and a distance a from one end (Fig. 23.15).

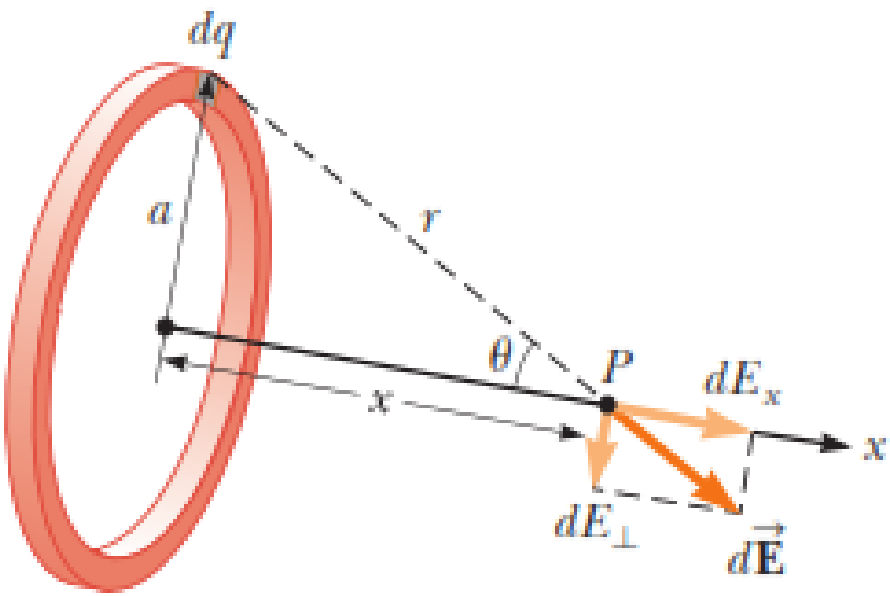


$$E = \frac{k_e Q}{a(\ell + a)}$$

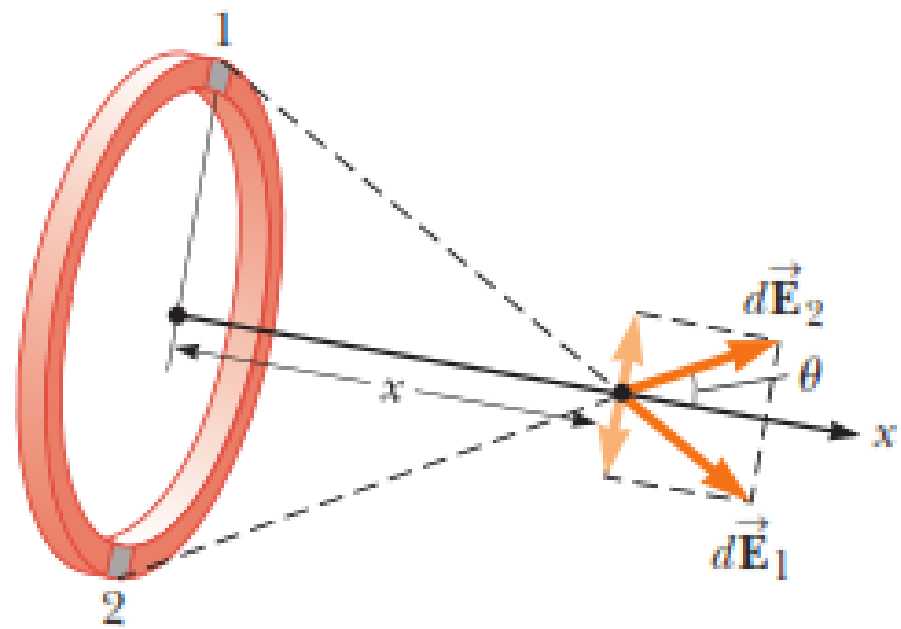
Example 23.8 The Electric Field of a Uniform Ring of Charge

A ring of radius a carries a uniformly distributed positive total charge Q. Calculate the electric field due to the ring at a point P lying a distance x from its center along the central axis perpendicular to the plane of the ring (Fig. 23.16a).

$$E = \frac{k_e x}{(a^2 + x^2)^{3/2}} Q$$



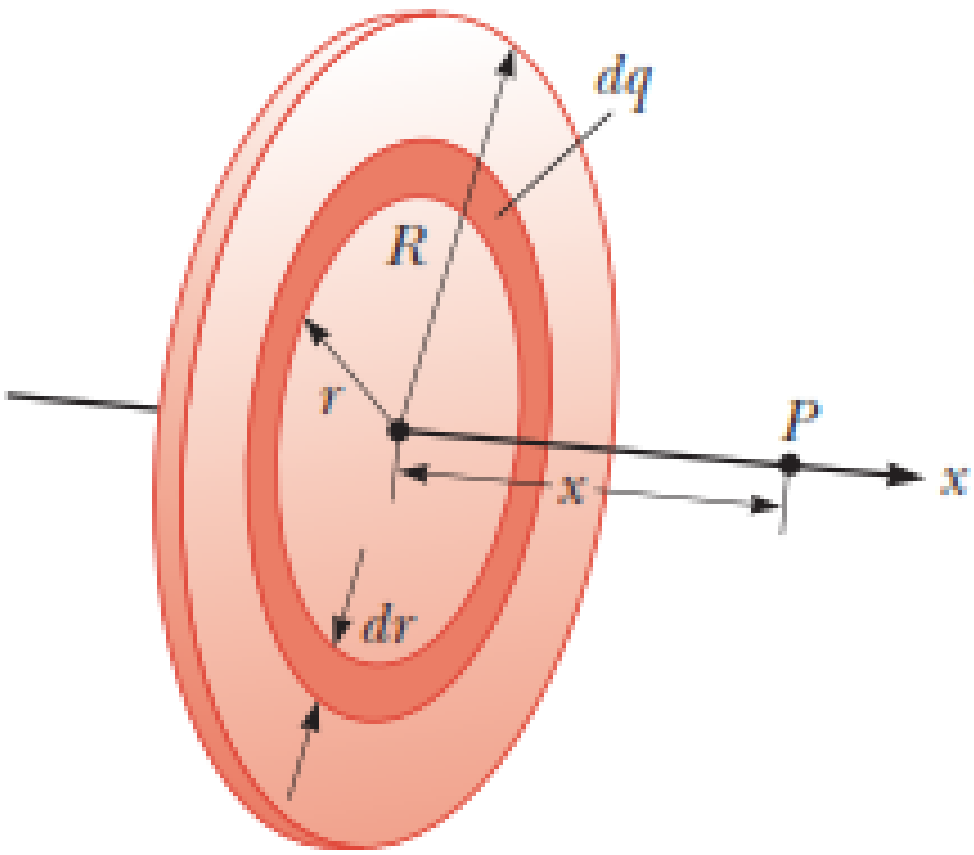
a



b

Example 23.9 The Electric Field of a Uniformly Charged Disk

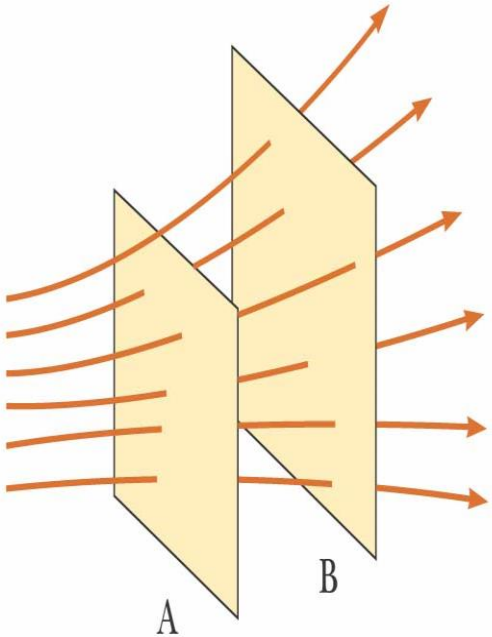
A disk of radius R has a uniform surface charge density σ. Calculate the electric field at a point P that lies along the central perpendicular axis of the disk and a distance x from the center of the disk (Fig. 23.17).



$$E_x = 2\pi k_e \sigma \left[1 - \frac{x}{(R^2 + x^2)^{1/2}} \right]$$

Sec 23.6: Electric Field Lines

- Field lines give us a means of representing the electric field pictorially
- The electric field vector $\vec{E}$ is tangent to the electric field line at each point
  - The line has a direction that is the same as that of the electric field vector
- The number of lines per unit area through a surface perpendicular to the lines is proportional to the magnitude of the electric field in that region
- The density of lines through surface A is greater than through surface B
- The magnitude of the electric field is greater on surface A than B
- The lines at different locations point in different directions
  - This indicates the field is nonuniform



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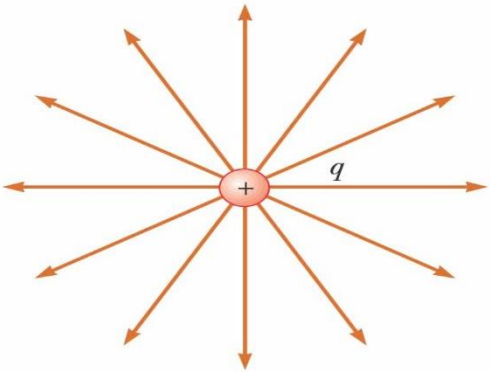
Electric Field Lines

- The field lines radiate outward in all directions
  - In three dimensions, the distribution is spherical
- The lines are directed away from the source charge
  - A positive test charge would be repelled away from the positive source charge



(a)

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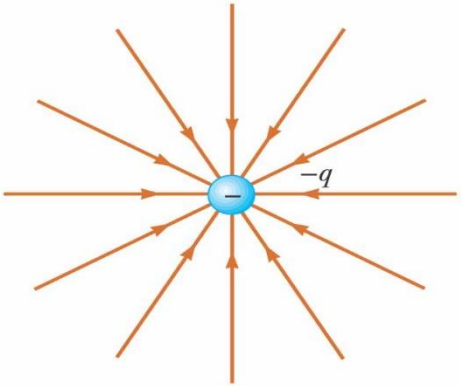
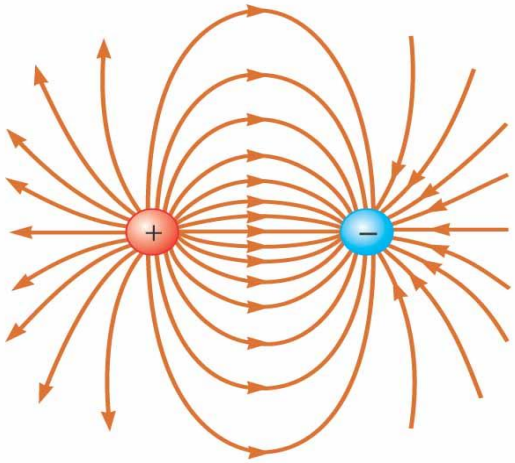
- The field lines radiate inward in all directions
- The lines are directed toward the source charge
  - A positive test charge would be attracted toward the negative source charge



- The charges are equal and opposite (Dipole)
- The number of field lines leaving the positive charge equals the number of lines terminating on the negative charge



(a)

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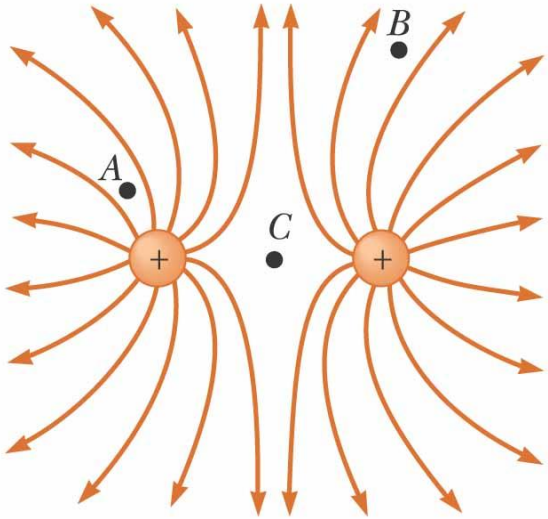
- The charges are equal and positive
- The same number of lines leave each charge since they are equal in magnitude
- At a great distance, the field is approximately equal to that of a single charge of $2q$



(a)

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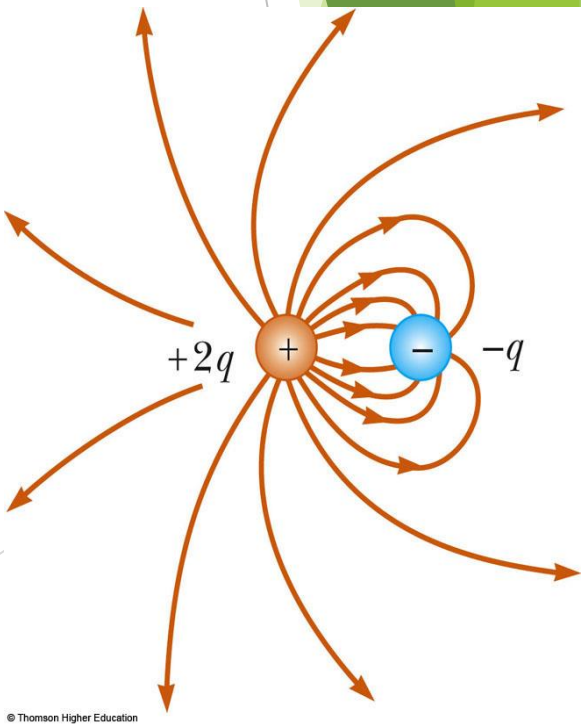
- The positive charge is twice the magnitude of the negative charge
- Two lines leave the positive charge for each line that terminates on the negative charge
- At a great distance, the field would be approximately the same as that due to a single charge of $+q$



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Electric Field Lines – Rules for Drawing

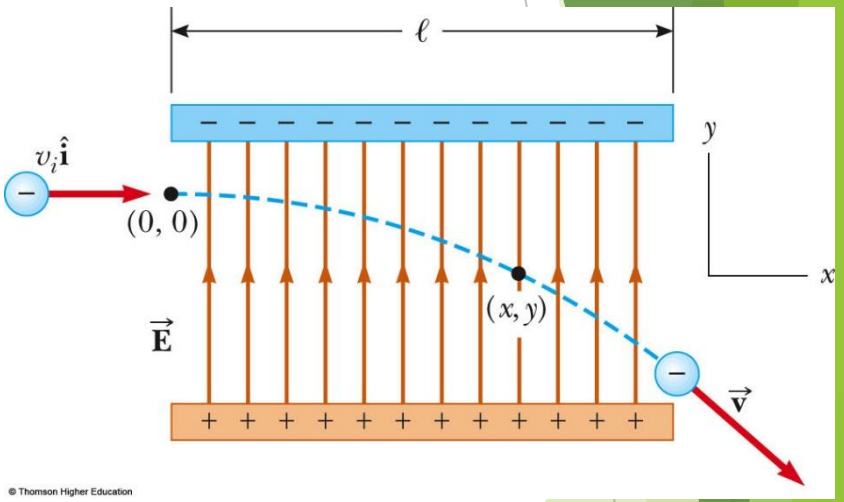
- The lines must begin on a positive charge and terminate on a negative charge
  - In the case of an excess of one type of charge, some lines will begin or end infinitely far away
- The number of lines drawn leaving a positive charge or approaching a negative charge is proportional to the magnitude of the charge
- No two field lines can cross
- Remember field lines are **not** material objects, they are a pictorial representation used to qualitatively describe the electric field

Sec 23.7: Motion of a Charged Particle in a Uniform Electric Field

- When a charged particle is placed in an electric field, it experiences an electrical force
- If this is the only force on the particle, it must be the net force
- The net force will cause the particle to accelerate according to Newton's second law
- a = q E / m $$\vec{\mathbf{F}}_e = q\vec{\mathbf{E}} = m\vec{\mathbf{a}}$$
- If $\vec{\mathbf{E}}$ is uniform, then the acceleration is constant
- If the particle has a positive charge, its acceleration is in the direction of the field
- If the particle has a negative charge, its acceleration is in the direction opposite the electric field
- Since the acceleration is constant, the kinematic equations can be used

Electron in a Uniform Field, Example

- The electron is projected horizontally into a uniform electric field
- The electron undergoes a downward acceleration
  - It is negative, so the acceleration is opposite the direction of the field
- Its motion is parabolic while between the plates

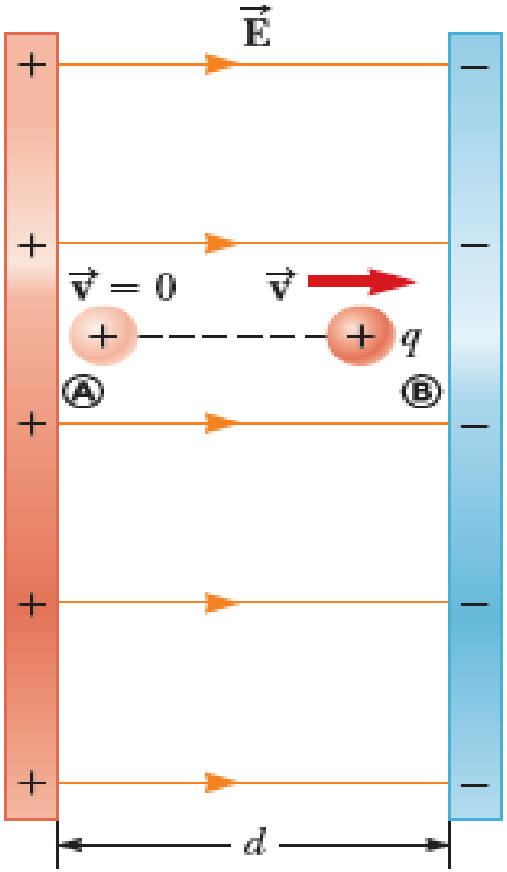


Example 23.10 An Accelerating Positive Charge: Two Models

AM

A uniform electric field $\vec{\mathbf{E}}$ is directed along the x axis between parallel plates of charge separated by a distance d as shown in Figure 23.23. A positive point charge q of mass m is released from rest at a point Ⓐ next to the positive plate and accelerates to a point Ⓑ next to the negative plate.

(A) Find the speed of the particle at Ⓑ by modeling it as a particle under constant acceleration.



$$v_f^2 = v_i^2 + 2a(x_f - x_i) = 0 + 2a(d - 0) = 2ad$$

$$v_f = \sqrt{2ad} = \sqrt{2\left(\frac{qE}{m}\right)d} = \boxed{\sqrt{\frac{2qEd}{m}}}$$

(B) Find the speed of the particle at Ⓑ by modeling it as a nonisolated system in terms of energy.

$$W = \Delta K$$

$$F_e\,\Delta x = K_{Ⓑ} - K_{Ⓐ} = \tfrac{1}{2}m v_f^2 - 0 \;\rightarrow\; v_f = \sqrt{\frac{2F_e\,\Delta x}{m}}$$

$$v_f = \sqrt{\frac{2(qE)(d)}{m}} = \boxed{\sqrt{\frac{2qEd}{m}}}$$